CT5102 Assignment 2 - Functions and Functionals (100 marks)

October 8, 2025

Table of contents

Objective

The aim of this assignment is to (1) create your own functions and (2) to use `lapply()` to process lists. Note it is recommended to use the docking station so that you may have a number of screens available to work on the assignment.

Write your own function to filter an atomic vector (25 marks)

Write a function `filter_numbers(v,n)` that takes two arguments

- An atomic vector of numbers (v)
- A whole number n (shoould be greater than 0)

The function should return those vector elements that are divisible by the inputted whole number, using the modulus operator `%%`.

An error check (with the `stop()` function) should be made to ensure the input number is greater than zero.

Here are test functions that can be used to check your results

```r
set.seed(100)
v <- sample(1:100,30)
v
```

```
 [1] 74 89 78 23 86 70  4 55 95  7 91 93 43 82 61 12 51 72 18 25  2 84 68 98 52
[26] 48 32 39 16 66
```

```r
filter_numbers(v,2)
```

```
 [1] 74 78 86 70  4 82 12 72 18  2 84 68 98 52 48 32 16 66
```

```r
filter_numbers(v,5)
```

```
[1] 70 55 95 25
```

```r
filter_numbers(v,10)
```

```
[1] 70
```

Write your own functional to process an atomic vector (25 marks)

Write a function `process_vector(v,fn)` that takes two arguments

- An atomic vector of numbers (v)
- A function that operates on the vector (e.g. min, max, mean, sd)

Here are test functions that can be used to check the results

```r
set.seed(100)
v <- sample(1:100,10)
v
```

```
 [1] 74 89 78 23 86 70  4 55 95  7
```

```r
process_vector(v,mean)
```

```
[1] 58.1
```

```r
process_vector(v,sd)
```

```
[1] 34.42367
```

Process the list `sw_films` (25 marks)

Here, using `lapply()`, process the list `sw_films` (from the library `repurrrsive`) to generate a simpler form of the list (`new_sw_films`) that only contains the:

- `episode_id`
- `title`
- `director`
- `release_date`

Within `lapply()` define an anonymous function that will be applied to each list element.

Here is the result you should get.

```r
str(new_sw_films)
```

```
List of 7
 $ :List of 4
  ..$ episode_id  : int 4
  ..$ title       : chr "A New Hope"
  ..$ director    : chr "George Lucas"
  ..$ release_date: chr "1977-05-25"
 $ :List of 4
  ..$ episode_id  : int 2
  ..$ title       : chr "Attack of the Clones"
  ..$ director    : chr "George Lucas"
  ..$ release_date: chr "2002-05-16"
 $ :List of 4
  ..$ episode_id  : int 1
  ..$ title       : chr "The Phantom Menace"
  ..$ director    : chr "George Lucas"
  ..$ release_date: chr "1999-05-19"
 $ :List of 4
  ..$ episode_id  : int 3
  ..$ title       : chr "Revenge of the Sith"
  ..$ director    : chr "George Lucas"
  ..$ release_date: chr "2005-05-19"
```

```
 $ :List of 4
  ..$ episode_id  : int 6
  ..$ title       : chr "Return of the Jedi"
  ..$ director    : chr "Richard Marquand"
  ..$ release_date: chr "1983-05-25"
 $ :List of 4
  ..$ episode_id  : int 5
  ..$ title       : chr "The Empire Strikes Back"
  ..$ director    : chr "Irvin Kershner"
  ..$ release_date: chr "1980-05-17"
 $ :List of 4
  ..$ episode_id  : int 7
  ..$ title       : chr "The Force Awakens"
  ..$ director    : chr "J. J. Abrams"
  ..$ release_date: chr "2015-12-11"
```

Next, find a way to name each list element to have the episode_id.

Here is the result you should get.

```r
str(new_sw_films)
```

```
List of 7
 $ episode_id_4:List of 4
  ..$ episode_id  : int 4
  ..$ title       : chr "A New Hope"
  ..$ director    : chr "George Lucas"
  ..$ release_date: chr "1977-05-25"
 $ episode_id_2:List of 4
  ..$ episode_id  : int 2
  ..$ title       : chr "Attack of the Clones"
  ..$ director    : chr "George Lucas"
  ..$ release_date: chr "2002-05-16"
 $ episode_id_1:List of 4
  ..$ episode_id  : int 1
  ..$ title       : chr "The Phantom Menace"
  ..$ director    : chr "George Lucas"
  ..$ release_date: chr "1999-05-19"
 $ episode_id_3:List of 4
  ..$ episode_id  : int 3
  ..$ title       : chr "Revenge of the Sith"
  ..$ director    : chr "George Lucas"
```

```
  ..$ release_date: chr "2005-05-19"
 $ episode_id_6:List of 4
  ..$ episode_id  : int 6
  ..$ title       : chr "Return of the Jedi"
  ..$ director    : chr "Richard Marquand"
  ..$ release_date: chr "1983-05-25"
 $ episode_id_5:List of 4
  ..$ episode_id  : int 5
  ..$ title       : chr "The Empire Strikes Back"
  ..$ director    : chr "Irvin Kershner"
  ..$ release_date: chr "1980-05-17"
 $ episode_id_7:List of 4
  ..$ episode_id  : int 7
  ..$ title       : chr "The Force Awakens"
  ..$ director    : chr "J. J. Abrams"
  ..$ release_date: chr "2015-12-11"
```

Get the movies for a director (25 marks)

Write a function `get_movies()` that will return an atomic vector of the movies directed by a director. Make use of `lapply()` inside the function, as well as functions such as `unlist()` and `is.na()`.

The function should accept:

- the list of movies (e.g. `sw_films` or `new_sw_films`)
- the target director

It should then return the result as a character (atomic) vector.

Here is the result

```r
mov1 <- get_movies(sw_films,"George Lucas")
mov1
```

```
[1] "A New Hope"            "Attack of the Clones" "The Phantom Menace"
[4] "Revenge of the Sith"
```

```r
mov2 <- get_movies(new_sw_films,"Richard Marquand")
mov2
```

```
        episode_id_6
"Return of the Jedi"
```